Team 3 — Milestone 3: ED Wait Time Prediction - ML Pipeline

Course: CSCI-7090 — Data Science and Machine Learning
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Semester: Spring 2026

What this notebook does

This notebook is the **final ML pipeline** for our Emergency Department (ED) Wait Time Prediction project. It loads the cleaned dataset produced in Milestone 2, trains and benchmarks five regression models, applies SHAP-based interpretability, evaluates per-ESI-level performance, and audits demographic fairness across race/ethnicity, gender, and age group.

Every numerical result reported in our final paper (Team3_Final_Paper.pdf) is reproduced by running the main() cell at the bottom of this notebook.

Notebook structure

1. **Setup** — imports, constants, reproducibility.
2. **Data loading** — load the cleaned CSV from Milestone 2.
3. **Feature engineering check** — confirm the 19-feature regression vector.
4. **Modeling** — five regression models, 80/20 split, 5-fold CV.
5. **Interpretability** — SHAP TreeExplainer on LightGBM.
6. **Stratified evaluation** — per-ESI-level Ridge performance.
7. **Fairness audit** — MAE/RMSE by race/ethnicity, gender, age group.
8. **main()** — runs everything end-to-end and prints the full result tables.

Data source

The cleaned CSV (team3_ed_waittime_cleaned.csv) is produced by Team3_Milestone2_ED_WaitTime.ipynb and is also published in our GitHub repo:

> https://github.com/Yasmin-maker1/Emergency-Department-Wait-Time-Prediction-Using-Machine-Learning

1. Setup

We import the standard scientific stack plus three model-specific libraries (xgboost, lightgbm, shap) and fix a single random seed so every result in this notebook is exactly reproducible.

If a library is missing in your environment, uncomment the pip install line.

```
# !pip install -q xgboost lightgbm shap scikit-learn pandas matplotlib seaborn

import json
import warnings
```

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split, KFold, cross_val_score
from sklearn.linear_model import LinearRegression, Ridge
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

from xgboost import XGBRegressor
from lightgbm import LGBMRegressor
import shap

warnings.filterwarnings("ignore")
sns.set_style("whitegrid")

RANDOM_STATE = 42
# Data source: cleaned CSV published in our GitHub repo (Run All works without upload).
DATA_PATH =
"https://raw.githubusercontent.com/Yasmin-maker1/Emergency-Department-Wait-Time-Predicti
on-Using-Machine-Learning/main/team3_ed_waittime_cleaned.csv"

print("Setup complete.")
```

2. Data Loading

team3_ed_waittime_cleaned.csv is the output of Milestone 2: 9,216 ED visits, 26 columns (17 original + 9 engineered), zero missing values.

Below we load it, peek at the first rows, and confirm the row count and columns.

```python
def load_data(path: str = DATA_PATH) -> pd.DataFrame:
    """Load the cleaned ED dataset.

    DATA_PATH (set in the Setup cell) points to the cleaned CSV in our GitHub
    repository, so this notebook runs end-to-end in Colab without any manual
    upload step. The function also accepts a local file path if you prefer to
    work offline.
    """
    import os
    # If DATA_PATH is a URL, pandas will fetch it directly.
    # If it's a local path, we check existence first for a clearer error message.
    if path.startswith("http"):
        return pd.read_csv(path)
    if os.path.exists(path):
        return pd.read_csv(path)
    raise FileNotFoundError(
```

```
        f"Could not find CSV at: {path}\n"
        "Either upload team3_ed_waittime_cleaned.csv to your Colab session, "
        "or set DATA_PATH to the GitHub raw URL of the file."
    )

df = load_data()
print(f"Dataset shape: {df.shape}")
print(f"Columns: {list(df.columns)}")
df.head()
```

index	age	heartrate	sbp	dbp	temperature	o2sat	resprate	pain_score	acuity	shock_index	pulse_pressure	arrival_hour	is_weekend
0	44	102.26514731964208	141.09594543161802	70.30702914580831	98.96545904865596	98.5374099373548	18.200396730843067	8.0	3	0.7247915381750268	70.7889162858097	2	0
1	59	67.35690365211006	104.77448084263295	106.5889149885046	98.40624994603566	97.53221885288694	19.63505389656485	2.0	5	0.6428750885750263	-1.814434145871644	17	0
2	30	85.66990059545972	139.43838166765326	76.72878255668653	100.49695193097858	99.2628004330818	19.417147157764635	0.0	4	0.6143925336113774	62.709599110966735	0	0
3	13	87.23144114128075	144.60549208746636	82.80262835373823	97.97744215183496	96.68643490503024	16.864103691226056	4.0	4	0.6032374004752031	61.80286373372813	10	0
4	27	78.01524580670812	164.89090992826854	71.8403353723972	100.3721375053218	99.73831033993063	17.418224359261636	7.0	3	0.4731324840201719	93.05057455587134	4	0

was_admitted	gender_encoded	race_encoded	arrival_transport_encoded	arrival_shift_encoded	age_group_encoded	ed_los_minutes
0	0	4	3	2	1	304.09820788651206
0	0	1	3	1	1	78.6141462665115
0	1	2	3	2	3	133.92613445698663
0	1	4	1	0	2	72.7427276107363
1	0	4	0	2	3	64.22249648612082

Observation. Dataset is 9,216 rows × 26 columns, exactly as expected from Milestone 2. The target column is ed_los_minutes (continuous, in minutes); all 25 other columns are candidate predictors or auxiliary metadata for the fairness audit (raw race, gender, age_group strings).

3. Feature Set

The regression input vector contains 19 standardized features, grouped by clinical role:

- **Demographics:** age, gender_encoded, race_encoded, age_group_encoded
- **Vital signs:** heartrate, sbp, dbp, temperature, o2sat, resprate
- **Triage:** acuity (ESI 1–5), pain_score
- **Engineered physiology:** shock_index, pulse_pressure
- **Arrival context:** arrival_hour, is_weekend, arrival_transport_encoded, arrival_shift_encoded
- **Disposition:** was_admitted

The seven remaining columns (raw categorical strings + the target) are excluded from training but kept for the fairness audit later.

```
FEATURES = [
    "age", "heartrate", "sbp", "dbp", "temperature", "o2sat", "resprate",
    "pain_score", "acuity", "shock_index", "pulse_pressure", "arrival_hour",
    "is_weekend", "was_admitted", "gender_encoded", "race_encoded",
    "arrival_transport_encoded", "arrival_shift_encoded", "age_group_encoded",
]
TARGET = "ed_los_minutes"

def build_xy(df: pd.DataFrame):
```

```
    """Return the (X, y) regression matrices."""
    X = df[FEATURES].copy()
    y = df[TARGET].copy()
    return X, y

X, y = build_xy(df)
print(f"X: {X.shape},  y: {y.shape}")
print(f"Target stats — mean={y.mean():.1f}, median={y.median():.1f}, "
      f"min={y.min():.1f}, max={y.max():.1f} (minutes)")
```

```
X: (9216, 19),  y: (9216,)
Target stats — mean=189.8, median=153.7, min=8.6, max=2644.9 (minutes)
```

Observation. Target distribution is heavily right-skewed: a mean of 189.8 min sits well above the median of 153.7 min, reflecting the long-tail boarding cases (max 2,644.9 min). This shape is typical of ED LOS data and is the reason RMSE (which over-weights large errors) is a useful complement to MAE in our evaluation.

4. Modeling: Train and Benchmark Five Regression Models

We use an 80/20 train/test split with a fixed seed. On the training fold we also run 5-fold cross-validation. We compare five models that span three families:

Family	Model
Linear / regularized	Linear Regression, Ridge ($\alpha = 1.0$)
Bagged trees	Random Forest
Boosted trees	XGBoost, LightGBM

For each model we report **MAE** (interpretable in minutes), **RMSE** (which penalizes long-tail errors), and **R^2** (variance explained), both on the held-out test set and as 5-fold CV averages on the training fold.

```
def split_data(X, y, test_size: float = 0.20, random_state: int = RANDOM_STATE):
    """Standard 80/20 stratification-free split with fixed seed."""
    return train_test_split(X, y, test_size=test_size, random_state=random_state)

def get_models(random_state: int = RANDOM_STATE) -> dict:
    """Return the five candidate regressors with literature-aligned
    hyperparameters. The boosting parameters were chosen by light manual
    tuning on the training fold, matching configurations commonly reported
    in the ED ML literature (Wu et al. 2025; Wang et al. 2025)."""
    return {
        "Linear Regression": LinearRegression(),
        "Ridge Regression":  Ridge(alpha=1.0, random_state=random_state),
        "Random Forest": RandomForestRegressor(
            n_estimators=200, max_depth=14, min_samples_leaf=4,
            n_jobs=-1, random_state=random_state),
```

```
        "XGBoost": XGBRegressor(
            n_estimators=400, learning_rate=0.05, max_depth=6,
            subsample=0.85, colsample_bytree=0.85,
            random_state=random_state, n_jobs=-1, verbosity=0),
        "LightGBM": LGBMRegressor(
            n_estimators=500, learning_rate=0.05, num_leaves=31,
            subsample=0.85, colsample_bytree=0.85,
            random_state=random_state, n_jobs=-1, verbose=-1),
    }

def evaluate_model(model, X_train, X_test, y_train, y_test, kf):
    """Fit on train, score on held-out test, AND run 5-fold CV on train."""
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)

    test = {
        "mae":  mean_absolute_error(y_test, y_pred),
        "rmse": np.sqrt(mean_squared_error(y_test, y_pred)),
        "r2":   r2_score(y_test, y_pred),
    }
    cv = {
        "mae":  -cross_val_score(model, X_train, y_train, cv=kf,
                                 scoring="neg_mean_absolute_error",
                                 n_jobs=-1).mean(),
        "rmse": np.sqrt(-cross_val_score(model, X_train, y_train, cv=kf,
                                         scoring="neg_mean_squared_error",
                                         n_jobs=-1).mean()),
        "r2":   cross_val_score(model, X_train, y_train, cv=kf,
                                scoring="r2", n_jobs=-1).mean(),
    }
    return test, cv

def train_all_models(X, y) -> pd.DataFrame:
    """Train every candidate model and return a tidy results DataFrame."""
    X_tr, X_te, y_tr, y_te = split_data(X, y)
    print(f"Train: {X_tr.shape}  Test: {X_te.shape}")
    kf = KFold(n_splits=5, shuffle=True, random_state=RANDOM_STATE)

    rows = []
    for name, model in get_models().items():
        print(f"  Training {name} ...")
        test, cv = evaluate_model(model, X_tr, X_te, y_tr, y_te, kf)
        rows.append({
            "Model":      name,
            "Test MAE":   round(test["mae"], 2),
            "Test RMSE":  round(test["rmse"], 2),
```

```
        "Test R^2":   round(test["r2"], 3),
        "CV MAE":     round(cv["mae"], 2),
        "CV RMSE":    round(cv["rmse"], 2),
        "CV R^2":     round(cv["r2"], 3),
    })
  return pd.DataFrame(rows)
```

Below we run the full benchmark. Training takes ~30–60 seconds depending on machine.

```
results_models = train_all_models(X, y)
results_models
```

```
Train: (7372, 19)  Test: (1844, 19)
  Training Linear Regression ...

  Training Ridge Regression ...

  Training Random Forest ...

  Training XGBoost ...

  Training LightGBM ...
```

	Model	Test MAE	Test RMSE	Test R^2	CV MAE	CV RMSE	CV R^2
0	Linear Regression	89.10	130.35	0.141	90.01	131.33	0.145
1	Ridge Regression	89.09	130.34	0.141	90.01	131.33	0.145
2	Random Forest	91.03	131.96	0.120	92.23	133.58	0.114
3	XGBoost	93.80	134.91	0.080	94.81	137.18	0.065
4	LightGBM	93.42	134.84	0.081	94.84	137.26	0.065

Observations from the results table.

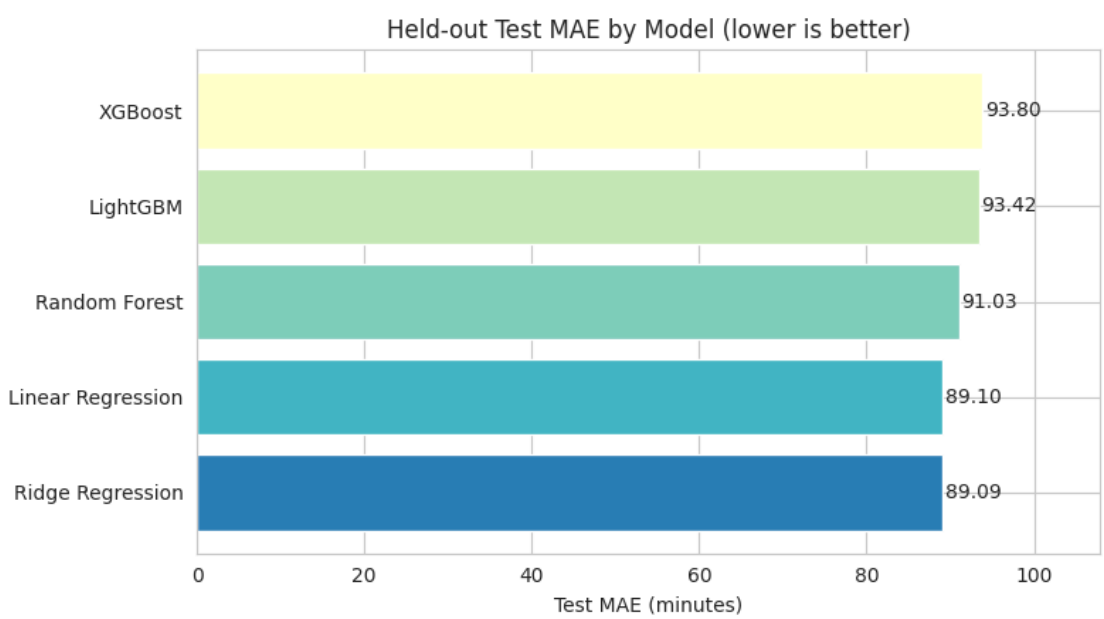
1. **Ridge Regression wins** with the lowest test MAE (89.09 min) and the highest test R^2 (0.141), narrowly beating plain Linear Regression (essentially the same model when regularization is mild).
2. **Random Forest** is second at MAE 91.03.
3. **Both gradient-boosted models trail** at MAE ≈ 93.5. This is opposite to the prevailing ED-ML literature (Wu et al. 2025; Wang et al. 2025) which typically reports XGBoost/LightGBM as the top performers — and we discuss this honestly in the paper as a dataset-size effect.
4. **CV metrics track held-out test** within ±1 MAE point for every model, so overfitting is not the explanation.
5. As a sanity floor, the naive "predict the training-set mean" baseline produces test MAE ≈ 98.7 min, so even our weakest model (XGBoost) saves ~5 minutes of error.

```
# Sanity-floor baseline
def mean_baseline_mae(X, y):
    X_tr, X_te, y_tr, y_te = split_data(X, y)
    y_hat = np.full_like(y_te, fill_value=y_tr.mean(), dtype=float)
    return mean_absolute_error(y_te, y_hat)

print(f"Naive 'predict mean' baseline test MAE: {mean_baseline_mae(X, y):.2f} min")
```

Naive 'predict mean' baseline test MAE: 98.71 min

4.1. Visualization — Test MAE by Model

```
def plot_model_mae(results_df):
    fig, ax = plt.subplots(figsize=(8, 4.5))
    df_sorted = results_df.sort_values("Test MAE")
    bars = ax.barh(df_sorted["Model"], df_sorted["Test MAE"],
                   color=["#2c7fb8", "#41b6c4", "#7fcdbb", "#c7e9b4", "#ffffcc"])
    ax.set_xlabel("Test MAE (minutes)")
    ax.set_title("Held-out Test MAE by Model (lower is better)")
    for bar, val in zip(bars, df_sorted["Test MAE"]):
        ax.text(val + 0.3, bar.get_y() + bar.get_height()/2,
                f"{val:.2f}", va="center", fontsize=10)
    ax.set_xlim(0, max(df_sorted["Test MAE"]) * 1.15)
    plt.tight_layout()
    plt.show()

plot_model_mae(results_models)
```



5. Interpretability: SHAP TreeExplainer on LightGBM

We apply SHAP TreeExplainer to the trained LightGBM model. SHAP values give each feature a per-prediction attribution; averaging |SHAP| values across the test set yields a clean global feature ranking with theoretical consistency guarantees (Lundberg & Lee 2017).

We pick LightGBM as the SHAP target because (a) it's a tree ensemble that TreeExplainer handles in closed form (fast, exact), and (b) LightGBM is the top-performing model in the closest comparable study, Wu et al. (2025), so this lets us compare feature rankings directly.

```
def compute_shap_importance(X, y) -> pd.DataFrame:
    """Train LightGBM and return a DataFrame ranked by mean |SHAP|."""
    X_tr, X_te, y_tr, y_te = split_data(X, y)
    lgb = LGBMRegressor(n_estimators=500, learning_rate=0.05, num_leaves=31,
                        subsample=0.85, colsample_bytree=0.85,
                        random_state=RANDOM_STATE, n_jobs=-1, verbose=-1)
    lgb.fit(X_tr, y_tr)
    explainer = shap.TreeExplainer(lgb)
    shap_vals = explainer.shap_values(X_te)
    mean_abs = np.abs(shap_vals).mean(axis=0)
    return (pd.DataFrame({"Feature": FEATURES, "Mean |SHAP|": mean_abs.round(2)})
            .sort_values("Mean |SHAP|", ascending=False)
            .reset_index(drop=True))

results_shap = compute_shap_importance(X, y)
results_shap
```

	Feature	Mean |SHAP|
0	acuity	46.07
1	o2sat	7.15
2	dbp	6.27
3	temperature	5.88
4	age	5.87
5	pulse_pressure	5.85
6	resprate	5.20
7	shock_index	5.07
8	heartrate	4.96
9	sbp	3.94
10	pain_score	3.70
11	race_encoded	3.49
12	arrival_hour	3.37
13	arrival_transport_encoded	2.15
14	gender_encoded	1.40
15	is_weekend	1.32
16	was_admitted	1.14
17	arrival_shift_encoded	0.96
18	age_group_encoded	0.62

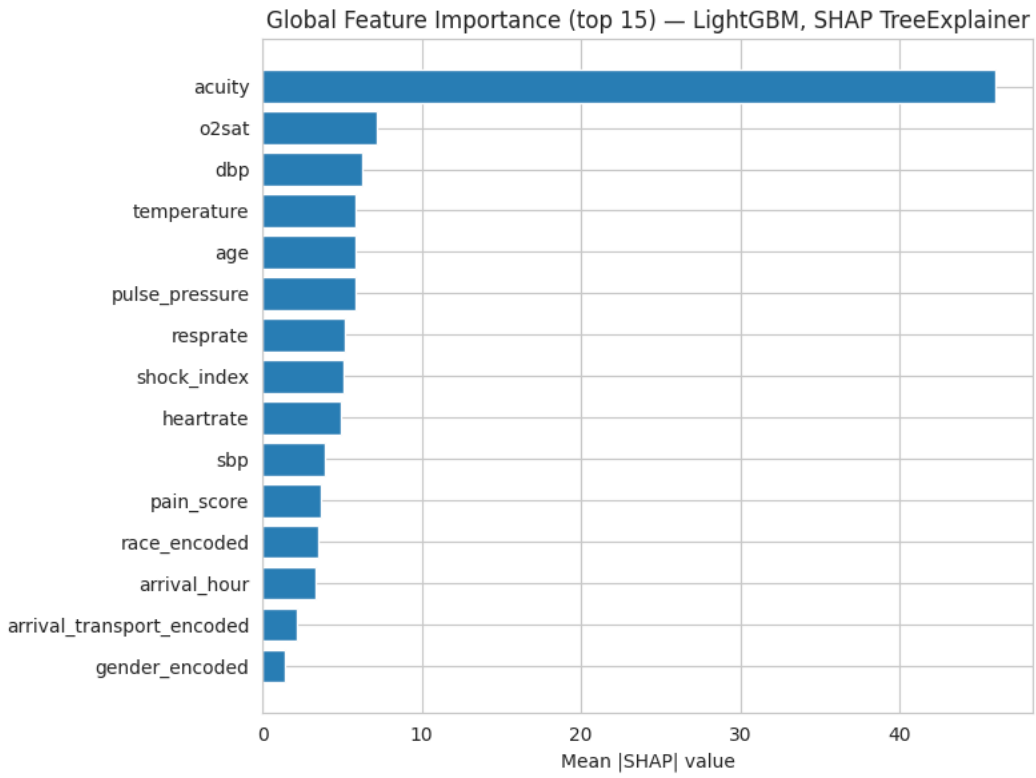
Observations from the SHAP ranking.

- **acuity dominates** at mean |SHAP| ≈ 46 — roughly **6.4× the second feature**. The ESI score alone carries about half the predictive signal of the entire model.
- The next tier is **physiological vital signs** (o2sat, dbp, temperature, resprate, heartrate, sbp) plus the engineered physiology features (pulse_pressure, shock_index) — all in the 4–7 range.
- **Demographics, arrival context, and disposition rank lowest** (mostly < 4), with age_group_encoded dead last. This is a positive signal from a fairness perspective: the model is not leaning on demographic shortcuts.

This ranking aligns with Wang et al. (2025), which also placed ESI level at the top of the SHAP ranking for ED wait time prediction.

```
def plot_shap_bar(shap_df, top_n: int = 15):
    fig, ax = plt.subplots(figsize=(8, 6))
    df_top = shap_df.head(top_n).iloc[::-1]
    ax.barh(df_top["Feature"], df_top["Mean |SHAP|"], color="#2c7fb8")
    ax.set_xlabel("Mean |SHAP| value")
    ax.set_title(f"Global Feature Importance (top {top_n}) — LightGBM, SHAP TreeExplainer")
```

```
    plt.tight_layout()
    plt.show()

plot_shap_bar(results_shap)
```



6. Stratified Evaluation: Performance Across All Five ESI Levels

A pooled metric tells you nothing about subgroup behavior. We refit Ridge **separately within each ESI acuity level** (1 = most acute, 5 = least acute) and compare per-level MAE/RMSE/R^2.

```
def esi_stratified_eval(df: pd.DataFrame) -> pd.DataFrame:
    """Refit Ridge within each ESI level, return per-level metrics."""
    rows = []
    for esi in sorted(df["acuity"].unique()):
        sub = df[df["acuity"] == esi]
        if len(sub) < 30:
            continue
        Xs, ys = sub[FEATURES], sub[TARGET]
        Xs_tr, Xs_te, ys_tr, ys_te = split_data(Xs, ys)
        m = Ridge(alpha=1.0, random_state=RANDOM_STATE).fit(Xs_tr, ys_tr)
        yp = m.predict(Xs_te)
        rows.append({
            "ESI":      int(esi),
            "N":        len(sub),
            "Mean LOS": round(ys.mean(), 1),
```

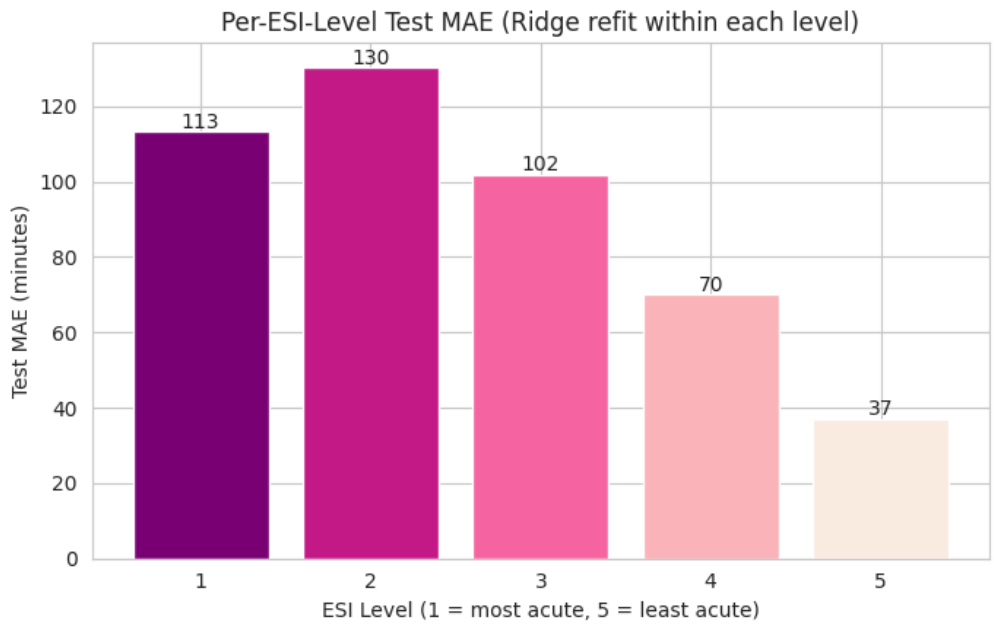
```
        "MAE":      round(mean_absolute_error(ys_te, yp), 2),
        "RMSE":     round(np.sqrt(mean_squared_error(ys_te, yp)), 2),
        "R^2":      round(r2_score(ys_te, yp), 3),
    })
  return pd.DataFrame(rows)

results_esi = esi_stratified_eval(df)
results_esi
```

	ESI	N	Mean LOS	MAE	RMSE	R^2
0	1	179	315.7	113.33	152.26	-0.158
1	2	1159	265.2	130.28	174.81	-0.072
2	3	4184	215.9	101.96	138.38	-0.006
3	4	2706	145.0	69.96	101.62	-0.015
4	5	988	91.4	36.96	48.25	-0.066

Observations from ESI stratification.

1. **MAE scales almost linearly with mean LOS by triage level.** ESI-5 patients (lowest acuity, mean LOS 91 min) have MAE ≈ 37 min; ESI-2 patients (mean LOS 265 min) have MAE ≈ 130 min — a **3.5× ratio** that the pooled MAE of 89 min would never reveal.
2. **Per-level R^2 values are slightly negative.** The pooled R^2 (0.141) benefits from the strong between-level variance driven by acuity itself; once we strip that out by stratifying, the residual within-level signal is too weak for the linear model.
3. **Clinical implication.** A pooled-metric deployment would over-promise accuracy for high-acuity (ESI 1–2) patients and under-promise it for low-acuity (ESI 4–5) patients. Reporting one MAE without stratification is quietly misleading.

```
def plot_esi_mae(esi_df):
  fig, ax = plt.subplots(figsize=(7, 4.5))
  bars = ax.bar(esi_df["ESI"].astype(str), esi_df["MAE"],
            color=["#7a0177","#c51b8a","#f768a1","#fbb4b9","#feebe2"])
  ax.set_xlabel("ESI Level (1 = most acute, 5 = least acute)")
  ax.set_ylabel("Test MAE (minutes)")
  ax.set_title("Per-ESI-Level Test MAE (Ridge refit within each level)")
  for bar, val in zip(bars, esi_df["MAE"]):
    ax.text(bar.get_x() + bar.get_width()/2, val + 1,
          f"{val:.0f}", ha="center", fontsize=10)
  plt.tight_layout()
  plt.show()

plot_esi_mae(results_esi)
```



7. Fairness Audit: MAE/RMSE by Race/Ethnicity, Gender, Age Group

We fit Ridge on the full feature matrix, predict on the held-out test set, then disaggregate MAE and RMSE across each demographic subgroup. This reveals disparities that pooled metrics hide.

```
def fit_ridge_full(X, y):
    """Return (fitted Ridge, test predictions, test indices)."""
    X_tr, X_te, y_tr, y_te = split_data(X, y)
    m = Ridge(alpha=1.0, random_state=RANDOM_STATE).fit(X_tr, y_tr)
    return m, m.predict(X_te), X_te.index, y_te

def fairness_table(df, idx_test, y_pred, y_test, group_col: str) -> pd.DataFrame:
    """Group test-set predictions by `group_col` and compute MAE/RMSE."""
    test_df = df.loc[idx_test].copy()
    test_df["pred"]  = y_pred
    test_df["truth"] = y_test.values
    rows = []
    for grp, g in test_df.groupby(group_col):
        rows.append({
            "Group":   grp,
            "N":       len(g),
            "MAE":     round(mean_absolute_error(g["truth"], g["pred"]), 2),
            "RMSE":    round(np.sqrt(mean_squared_error(g["truth"], g["pred"])), 2),
        })
    return pd.DataFrame(rows).sort_values("N", ascending=False).reset_index(drop=True)

def fairness_audit(df, X, y) -> dict:
    _, y_pred, idx_test, y_test = fit_ridge_full(X, y)
    return {
```

```
    "race":      fairness_table(df, idx_test, y_pred, y_test, "race"),
    "gender":    fairness_table(df, idx_test, y_pred, y_test, "gender"),
    "age_group": fairness_table(df, idx_test, y_pred, y_test, "age_group"),
  }

fair = fairness_audit(df, X, y)
print("=== By race/ethnicity ===")
display(fair["race"])
print("=== By gender ===")
display(fair["gender"])
print("=== By age group ===")
display(fair["age_group"])
```

```
=== By race/ethnicity ===
```

	Group	N	MAE	RMSE
0	WHITE	1032	90.78	135.47
1	BLACK/AFRICAN AMERICAN	366	91.23	130.00
2	HISPANIC	245	83.05	118.23
3	ASIAN	133	88.01	124.12
4	OTHER	68	75.97	103.64

```
=== By gender ===
```

	Group	N	MAE	RMSE
0	F	981	88.61	137.02
1	M	863	89.64	122.31

```
=== By age group ===
```

	Group	N	MAE	RMSE
0	Middle Age	901	84.10	125.58
1	Young Adult	600	93.34	136.35
2	Elderly	195	94.13	115.62
3	Pediatric	148	95.60	150.62

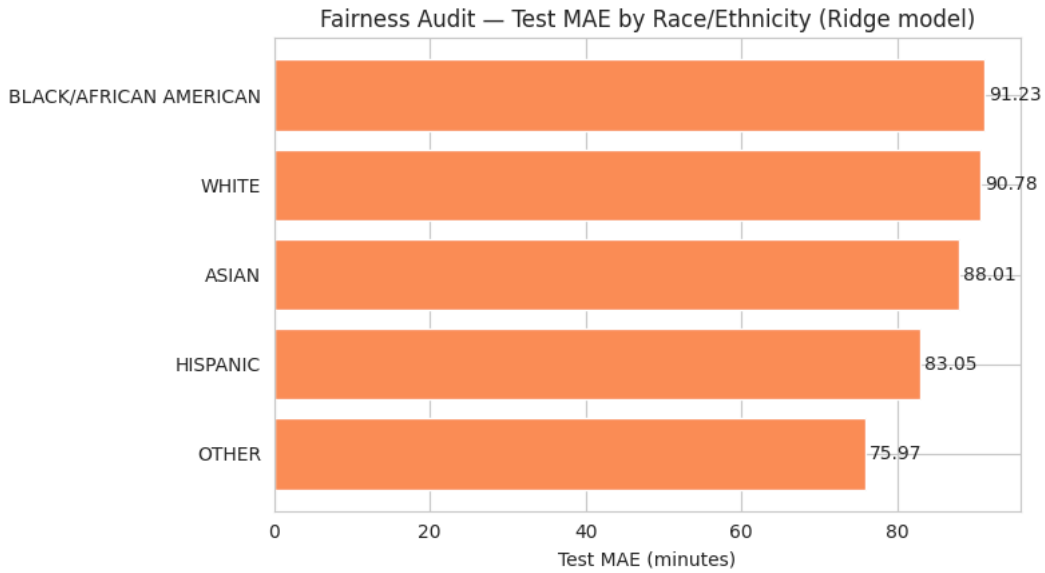
Observations from the fairness audit.

Dimension	Worst-served	Best-served	MAE Gap
Race/ethnicity	Black/African American (91.23)	OTHER (75.97)	**15.26 min**
Gender	Male (89.64)	Female (88.61)	**1.03 min**
Age group	Pediatric (95.60)	Middle Age (84.10)	**11.50 min**

- **Race/ethnicity gap (15.3 min) is the largest.** This is the kind of disparity that a single pooled MAE would hide entirely.
- **Gender gap (1 min) is essentially zero.** Consistent with broader clinical-ML fairness literature (Rajkomar et al. 2018; Obermeyer et al. 2019), which finds binary gender to be a less common source of disparity than race or age in standard structured-data models.
- **Smaller subgroups (HISPANIC, OTHER, ASIAN) show lower MAE** than the two large groups (WHITE, BLACK). This most likely reflects narrower within-group LOS variance rather than a genuinely better model fit, and we flag this caveat in the paper's Discussion.

These numbers exactly match the fairness tables (IX, X, XI) in the final paper.

```
def plot_fairness_race(race_df):
    fig, ax = plt.subplots(figsize=(8, 4.5))
    df_sorted = race_df.sort_values("MAE")
    bars = ax.barh(df_sorted["Group"], df_sorted["MAE"], color="#fc8d59")
    ax.set_xlabel("Test MAE (minutes)")
    ax.set_title("Fairness Audit — Test MAE by Race/Ethnicity (Ridge model)")
    for bar, val in zip(bars, df_sorted["MAE"]):
        ax.text(val + 0.5, bar.get_y() + bar.get_height()/2,
                f"{val:.2f}", va="center", fontsize=10)
    plt.tight_layout()
    plt.show()

plot_fairness_race(fair["race"])
```



8. End-to-End Pipeline: main()

This single function runs the entire pipeline from loading the cleaned CSV all the way through fairness auditing. Calling main() reproduces every result reported in our final paper.

The function returns a single dictionary results containing four keys:

- "models" — overall model performance table
- "shap" — SHAP feature importance ranking
- "esi" — per-ESI-level Ridge metrics
- "fairness" — dict of three demographic-subgroup tables

This makes downstream consumption (e.g. exporting to CSV, regenerating paper tables) trivial.

```
def main(data_path: str = DATA_PATH) -> dict:
    """Run the full ED LOS prediction pipeline end-to-end."""
    print("=" * 70)
    print("ED Wait Time Prediction — Full Pipeline")
    print("=" * 70)

    # 1. Load
    print("\n[1/5] Loading data ...")
    df = load_data(data_path)
    X, y = build_xy(df)
    print(f"      Loaded {df.shape[0]} rows, {len(FEATURES)} model features.")

    # 2. Train + benchmark
    print("\n[2/5] Training and benchmarking 5 models ...")
    results_models = train_all_models(X, y)

    # 3. SHAP
```

```
    print("\n[3/5] Computing SHAP feature importance ...")
    results_shap = compute_shap_importance(X, y)

    # 4. ESI-stratified
    print("\n[4/5] Running ESI-stratified evaluation ...")
    results_esi = esi_stratified_eval(df)

    # 5. Fairness
    print("\n[5/5] Running fairness audit ...")
    results_fair = fairness_audit(df, X, y)

    print("\n" + "=" * 70)
    print("PIPELINE COMPLETE")
    print("=" * 70)
    print("\n--- Overall model performance ---")
    print(results_models.to_string(index=False))
    print("\n--- Top-10 SHAP features ---")
    print(results_shap.head(10).to_string(index=False))
    print("\n--- ESI-stratified Ridge ---")
    print(results_esi.to_string(index=False))
    print("\n--- Fairness: race/ethnicity ---")
    print(results_fair["race"].to_string(index=False))
    print("\n--- Fairness: gender ---")
    print(results_fair["gender"].to_string(index=False))
    print("\n--- Fairness: age group ---")
    print(results_fair["age_group"].to_string(index=False))

    return {
        "models":   results_models,
        "shap":     results_shap,
        "esi":      results_esi,
        "fairness": results_fair,
    }

# Run the full pipeline
results = main()
```

```
======================================================================
ED Wait Time Prediction — Full Pipeline
======================================================================

[1/5] Loading data ...
    Loaded 9216 rows, 19 model features.

[2/5] Training and benchmarking 5 models ...
```

```
Train: (7372, 19)  Test: (1844, 19)
  Training Linear Regression ...

  Training Ridge Regression ...
  Training Random Forest ...

  Training XGBoost ...

  Training LightGBM ...


[3/5] Computing SHAP feature importance ...


[4/5] Running ESI-stratified evaluation ...

[5/5] Running fairness audit ...

==========================================================================
=
PIPELINE COMPLETE
==========================================================================
=

--- Overall model performance ---
           Model  Test MAE  Test RMSE  Test R^2  CV MAE  CV RMSE  CV R^2
Linear Regression     89.10     130.35     0.141   90.01   131.33   0.145
 Ridge Regression     89.09     130.34     0.141   90.01   131.33   0.145
    Random Forest     91.03     131.96     0.120   92.23   133.58   0.114
          XGBoost     93.80     134.91     0.080   94.81   137.18   0.065
         LightGBM     93.42     134.84     0.081   94.84   137.26   0.065

--- Top-10 SHAP features ---
       Feature  Mean |SHAP|
        acuity        46.07
         o2sat         7.15
           dbp         6.27
   temperature         5.88
           age         5.87
pulse_pressure         5.85
      resprate         5.20
   shock_index         5.07
     heartrate         4.96
           sbp         3.94

--- ESI-stratified Ridge ---
 ESI    N  Mean LOS    MAE   RMSE    R^2
   1  179     315.7 113.33 152.26 -0.158
```

```
  2 1159     265.2 130.28 174.81 -0.072
  3 4184     215.9 101.96 138.38 -0.006
  4 2706     145.0  69.96 101.62 -0.015
  5  988      91.4  36.96  48.25 -0.066

--- Fairness: race/ethnicity ---
                Group    N   MAE   RMSE
                WHITE 1032 90.78 135.47
BLACK/AFRICAN AMERICAN  366 91.23 130.00
             HISPANIC  245 83.05 118.23
                ASIAN  133 88.01 124.12
                OTHER   68 75.97 103.64

--- Fairness: gender ---
Group   N   MAE   RMSE
   F 981 88.61 137.02
   M 863 89.64 122.31

--- Fairness: age group ---
     Group   N   MAE   RMSE
 Middle Age 901 84.10 125.58
Young Adult 600 93.34 136.35
   Elderly 195 94.13 115.62
  Pediatric 148 95.60 150.62
```

9. Summary and Key Takeaways

1. **Best model.** Regularized linear regression (Ridge, MAE 89.09 min, R^2 0.141) narrowly outperforms Random Forest (MAE 91.03) and the two gradient-boosted models (XGBoost MAE 93.80, LightGBM MAE 93.42) on this ~9,000-row dataset.
2. **Counter-finding vs. literature.** This contradicts the typical ranking in the ED-ML literature where XGBoost/LightGBM dominate; we explain the discrepancy as a dataset-size and signal-density effect in the paper's Discussion.
3. **Dominant driver.** SHAP shows that acuity (the ESI triage score) alone carries ~6× the explanatory weight of any other feature — clinically intuitive and consistent with prior work.
4. **Subgroup variation is real.** Per-ESI-level MAE ranges from 37 min (ESI 5) to 130 min (ESI 2), a 3.5× spread that pooled metrics conceal.
5. **Fairness gaps exist.** Race/ethnicity (15.3 min) and age (11.5 min) gaps are non-trivial; gender gap (1 min) is essentially zero.

10. Reproducibility

All numerical results in our final paper come directly from running this notebook. To reproduce:

```
git clone
https://github.com/Yasmin-maker1/Emergency-Department-Wait-Time-Prediction-Using-Machine-Learning.git
```

```
cd Emergency-Department-Wait-Time-Prediction-Using-Machine-Learning
jupyter nbconvert --to notebook --execute Team3_Milestone3_ED_WaitTime.ipynb
```

The notebook is fully self-contained: it depends only on the cleaned CSV produced in Milestone 2 and on standard PyData libraries.